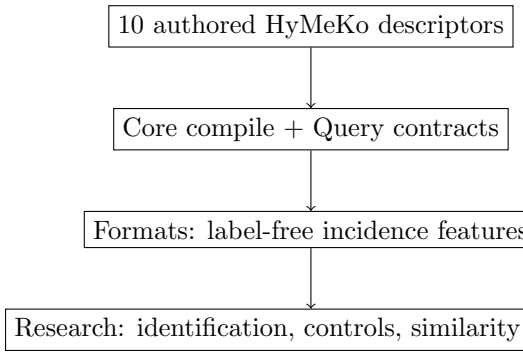
Ten-method structural identification test

Created-at: 2026-10-07T23:10+02:00 (Europe/Budapest). ETA: 45–60 minutes for the bounded corpus, owner API, controls and Amaterasu execution.

Scope Ten source-cited macro descriptions: DE, GA, PSO, HHO, GWO, WOA, BA, FA, CS, ABC. Compare typed read/write support and mechanism-enriched incidence neighborhoods. Blind labels and declaration names.



This tests projections, not ten optimizer implementations or numerical superiority.

Ownership Core retains compilation, canonical identity and limits; Query provides typed selection and dependencies; Formats extends its existing metaheuristic contract projection and CLI. Hystl research consumes exports for multiset comparison and presentation, with no source parser or graph engine. Existing DE/GA execution profiles remain frozen.

Files Transformation: hymeko_formats/src/metaheuristic/mod.rs, new catalog.rs; src/bin/metaheuristic_contract.rs; data/metaheuristics/catalog/de,ga,pso,hho,gwo,woa,ba,fa,cs,abc.hymeko and README.md; tests/metaheuristic_catalog.rs; README.md and generated SYMBOLS.md. Hystl: research/metaheuristics-pilot/identification_study.py, test_identification_st README.md. This plan and the report artifacts.

Interfaces Add a descriptive-catalog projection over CompiledModel and a structural mode to the existing contract CLI. Reuse exact-type Query checks and canonical hashing. Features contain only admitted semantic categories and incidence neighborhoods, never names, citations, paths, raw IDs or hashes. Canonical hashes remain provenance only. No dependencies or protected items changed; no simulator/renderer work.

Validation Preserve frozen DE/GA export tests; ten corpus inputs compile and roundtrip; reject missing/mixed/foreign roles and duplicate dependencies. Metamorphic rename/metadata/reorder controls, unknown and duplicate-prototype abstention; demonstrate same-support arithmetic counterexamples. Matrix output and plots on Amaterasu using installed matplotlib. No physics run.

Bounds Inherit 128 KiB source and 4096 declarations. One method per description; max 64 states, 64 operators and 128 role edges. Corpus exactly ten; bounded variants, 20 s subprocess timeout and fresh output directories. Two compiler workers, source test target under 1 GiB expected, no inference. Worst case input hits declared caps; fail rather than truncate.

Risks Hand-authored descriptors and deterministic transformations are a closed-corpus test, not held-out scientific identification. Formula strings are audit content, not executable semantics; collisions are reported. Structural similarity cannot prove equivalent stochastic behavior or runtime performance. Ambiguous paper semantics remain explicit. No new canonical graph or application-side parser.

Rollback Captured pre-edit bytes plus scoped hunks, no reset/stash or overwritten results. Existing dirty files preserved; no commits/pushes or new worktrees.